PCIE-1824 User Interface Manual

(Edition3.1)

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Welcome to use PCIE-1824 Driver User Manual. You can enter different sections by clicking the following links or you can use the table of contents to search and scan the content by index.

- Device Introduction
- Device Installation
- Device Setup
- Device Test
- Device Examples
- DAQNavi SDK Programming

These steps guide you how to install, configure and operate the PCIE-1824 device properly.

Step1: Power off the computer before installing the PCI/PCIE device. This step is no necessary for USB device.

Step2: According to Signal Connections or Switch and Jumper, adjust device switches and jumpers, then install the wiring cable and board.

Step3: Insert the PCI/PCIE device into a PCI/PCIE system slot or plug your USB module into the selected USB port.

Step4: Power on the computer. This step is no necessary for USB device.

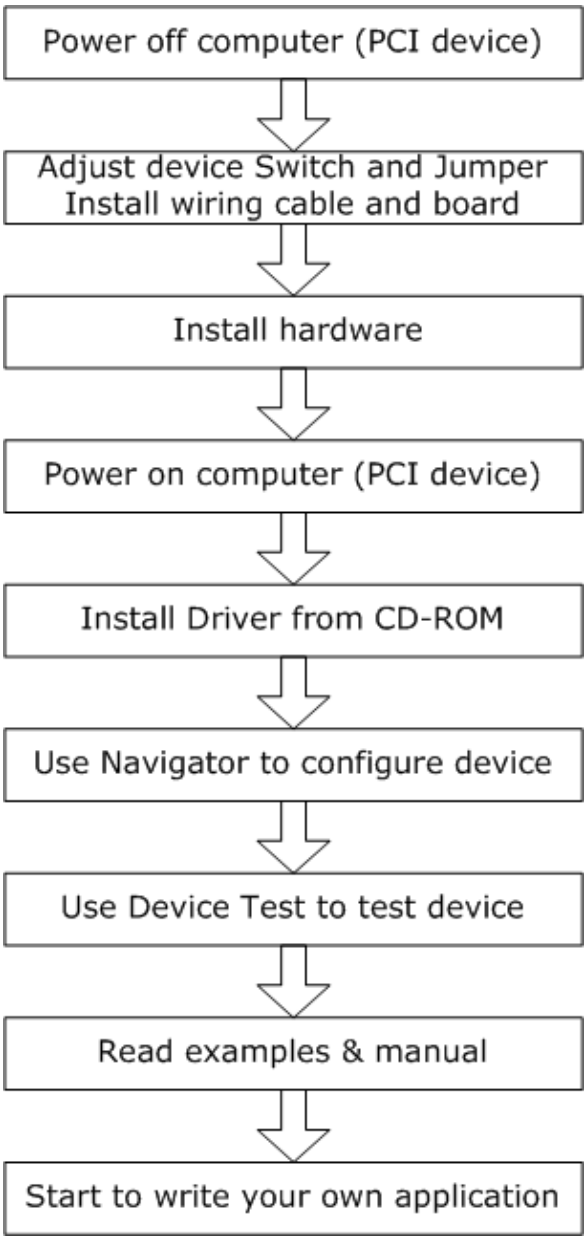
Step5: Insert the companion CD-ROM and install the corresponding DAQNavi driver and DAQNavi SDK.

Step6: Configure the Device Settings by using System Device Manager or Advantech Navigator.

Step7: Test and run device function by Device Test.

Step8: Refer to DAQNavi SDK examples and manual, you can get step-by-step instructions on how to build your own applications using DAQNavi SDK.

Step9: Programming with DAQNavi SDK to develop your own application.

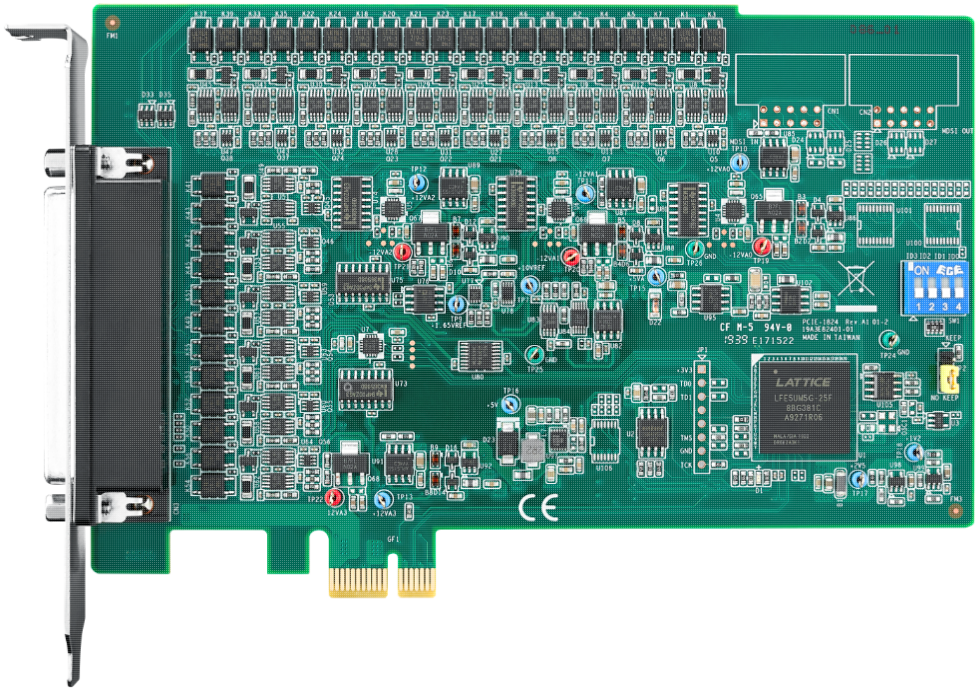

Power off computer (PCI device)
Adjust device Switch and Jumper
Install wiring cable and board
Install hardware
Power on computer (PCI device)
Install Driver from CD-ROM
Use Navigator to configure device
Use Device Test to test device
Read examples & manual
Start to write your own application

Introduction

The PCIE-1824 is a high-density multiple channel analog card for the PCIE bus where each analog output channel is equipped with a 16-bit DAC. It features optional voltages, current output and a Board ID switch. PCIE-1824 is an ideal solution for industrial applications where multiple analog output channels are required.

PCIE-1824

16-bit, 32-ch high-density multiple channel analog output card for the PCIE bus



Main Features

â¢ 32 high-density analog output channels

â¢ Flexible Output Range: Â±10 V, 0 ~ 20 mA and 4 ~ 20 mA

â¢ Synchronized output function

â¢ Keep output values when hot system reset

â¢ High ESD protection (2,000 VDC)

â¢ Board ID switch

Signal Connections include:

- Pin Assignment
- I/O Connector Signal Description
- Analog Output Connections
- Field Wiring Considerations

1. Pin Assignment

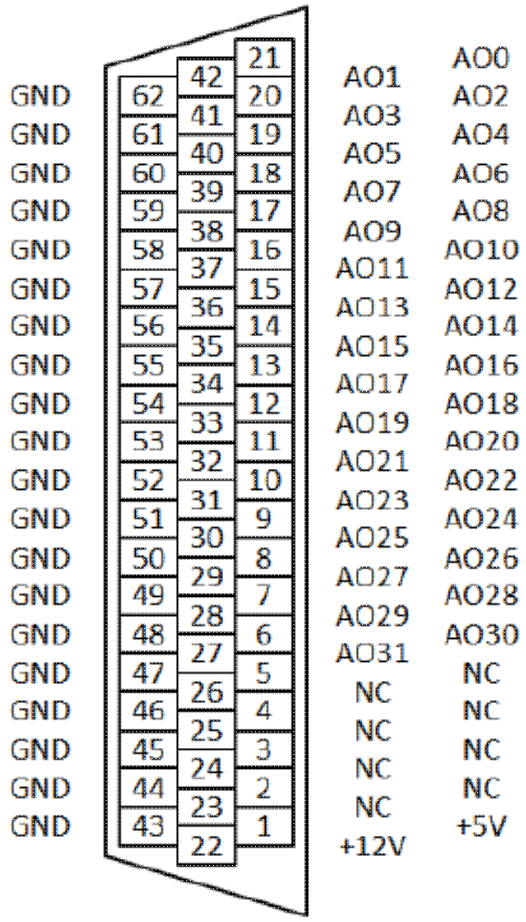


Figure 1-1: PCIE-1824 Pin Assignments and Connector Signal Descriptions

2. I/O Connector Signal Description

Pin Name	Type	Pin#	Description
		Analog Output	
AO0	O	21	Analog output channel 0
AO1	O	42	Analog output channel 1
AO2	O	20	Analog output channel 2
AO3	O	41	Analog output channel 3
AO4	O	19	Analog output channel 4
AO5	O	40	Analog output channel 5
AO6	O	18	Analog output channel 6
AO7	O	39	Analog output channel 7
AO8	O	17	Analog output channel 8
AO9	O	38	Analog output channel 9
AO10	O	16	Analog output channel 10
AO11	O	37	Analog output channel 11
AO12	O	15	Analog output channel 12
AO13	O	36	Analog output channel 13
AO14	O	14	Analog output channel 14
AO15	O	35	Analog output channel 15
AO16	O	13	Analog output channel 16
AO17	O	34	Analog output channel 17
AO18	O	12	Analog output channel 18
AO19	O	33	Analog output channel 19
AO20	O	11	Analog output channel 20
AO21	O	32	Analog output channel 21
AO22	O	10	Analog output channel 22
AO23	O	31	Analog output channel 23
AO24	O	9	Analog output channel 24
AO25	O	30	Analog output channel 25
AO26	O	8	Analog output channel 26
AO27	O	29	Analog output channel 27
AO28	O	7	Analog output channel 28
AO29	O	28	Analog output channel 29
AO30	O	6	Analog output channel 30
AO31	O	27	Analog output channel 31
		Power and Ground	
+12V	-	22	+12 V power supply for external use
+5V	-	1	+5 V power supply for external use
GND	-	43 ~ 63	Ground
		Others	
NC	-	2 ~ 5, 23 ~ 26	No connect.

Table 2-1: I/O Connector Signal Description

3. Analog Output Connections

The PCIE-1824 provides 32 channels of analog output (AO) signal generation with output ranges of Â±10 V, 0 ~ 20 mA, and 4 ~ 20 mA. The output range of each channel can be configured independently by software.

3.1 Voltage Output

For voltage output range (Â±10 V), the output configuration is single-ended. Signal connection is shown in Figure 3.3, and the minimum load resistance (RL) is 1 kÎ©.

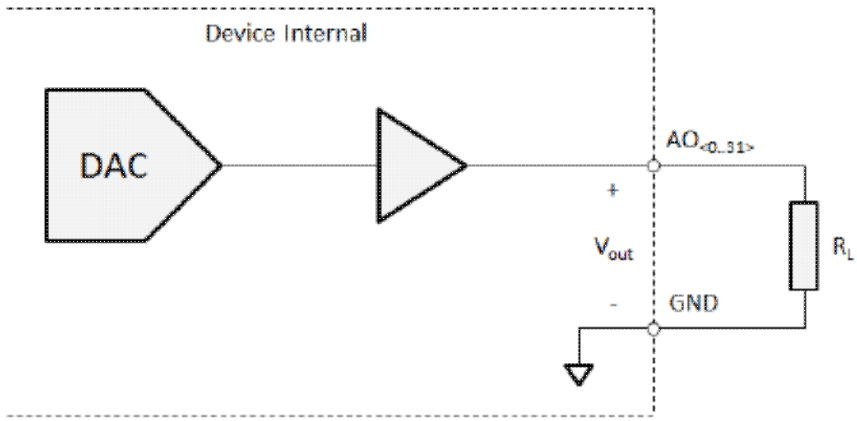


Figure 3-1: Voltage output signal connections

3.2 Current Output

For current output ranges (0 ~ 20 mA and 4 ~ 20 mA), current flows into the device (sink), and user must provide external power supply to the load. Signal connection is shown in the picture below.

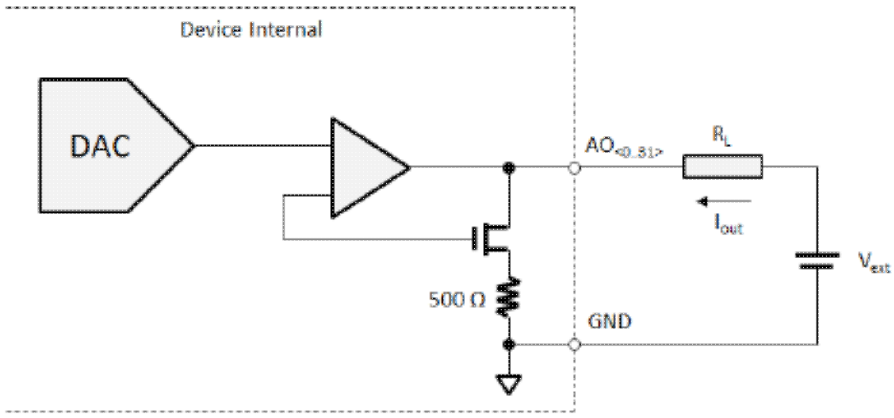


Figure 3-2: Current output signal connection

The maximum allowable load resistance (RL) can be calculated as follows. Vext is the external power supply in volts, which must not exceed 30 V.

$$R_{L,max} = \frac{V_{ext}}{20\,mA} - 0.5\,k\Omega$$

4. Field Wiring Considerations

When using a data acquisition devices to acquire signals from or generate signals to the outside world, noises and electromagnetic interferences from the outside world may significantly affect the results if precautions are not taken. The following instructions may be helpful to reduce these unwanted effects. The signal cables must be kept away from strong electromagnetic sources such as power lines, large electric motors, circuit breakers, and welding machines, since they may cause strong electromagnetic interference. Keep the analog signal cables away from any video monitor, since it can significantly affect data acquisition system. If the cable needs to unavoidably go through an area with significant electromagnetic interference, a shielded cable with twisted-pair wires should be used. Try to place the cable at a right angle to the nearby power lines to minimize any undesirable effects.

The Advantech PCIE-1824 is a Plug-and-Play device, which fully complies with the PCIE Specification. During card installation, all bus-related configurations such as base I/O address and interrupts are conveniently taken care of by the Plug-and-Play function. You have virtually no need to set any jumpers or DIP switches.

1. Jumper Settings

The default configuration after power on and hardware reset is to set all the analog input and analog output channels to open status (output voltage equals zero) so that external devices will not be damaged when the system starts or resets. When the system is hot reset, then the status of isolated digital output channels are selected by jumper JP2. Table 1-1 shows the possible configurations of jumper JP2.

JP2	Description
(1:2)	Keep last status after hot reset
(2:3)	Not keep last status after hot reset

Table 1-1: Card Jumper

2. Setting the BoardID Switch (SW1)

The PCIE-1824 has a built-in DIP switch (SW1), which is used to define each cardâ s board ID. When there are multiple cards on the same chassis, this board ID switch is useful for identifying each cardâ s device number.
After setting each PCIE-1824, you can identify each card in system with different device numbers. The default value of board ID is 0 and if you need to adjust it to other value, please set the SW1 by referring to Table 2-1.

SW1	Position 1	Position 2	Position 3	Position 4
BoardID	Bit0	Bit1	Bit2	Bit3
0	ON	ON	ON	ON
1	ON	ON	ON	OFF
2	ON	ON	OFF	ON
3	ON	ON	OFF	OFF
4	ON	OFF	ON	OFF
5	ON	OFF	ON	OFF
6	ON	OFF	OFF	ON
7	ON	OFF	OFF	OFF
8	OFF	ON	ON	ON
9	OFF	ON	ON	OFF
10	OFF	ON	OFF	ON
11	OFF	ON	OFF	OFF
12	OFF	OFF	ON	ON
13	OFF	OFF	ON	OFF
14	OFF	OFF	OFF	ON
15	OFF	OFF	OFF	OFF

Default Setting is 0

Table 2-1: Board ID Settings

Analog Output:

Channels	32, can be enabled/disabled each channel independently by software
Output configuration	Single-ended
Output range	±10 V/0 ~ 20 mA/4 ~ 20 mA (sink), software selectable per channel
D/A converter (DAC) resolution	16 bits
Update rates	Static update
Minimum load for voltage output	1 kΩ
Maximum external power for current output	30 V
Internal resistor	500 Ω
Power-up glitch removal	Yes
Auto-calibration	Yes

Voltage Output Accuracy:

Gain error	
Operating temperature within 5 °C of last auto-calibration temperature	< ±0.01%
Over full operating temperature range	< ±0.03%
Offset error	
Operating temperature within 5 °C of last auto-calibration temperature	< ±1 mV
Over full operating temperature range	< ±3 mV

Current Output Accuracy:

Gain error	
Operating temperature within 5 °C of last auto-calibration temperature	<±0.05%
Over full operating temperature range	<±0.15%
Offset error	
Operating temperature within 5 °C of last auto-calibration temperature	<±2.5 μA
Over full operating temperature range	<±10 μA

Dynamic Performance:

Slew rate	0.7 V/μs
Output noise	0.2 mVRMS
Channel to channel DC crosstalk	0.22 mV, measured channel at midscale, adjacent channel at full scale
Settling time	100 μs

Power Output:

+5 V	200 mA max.
+12 V	100 mA max.

General:

I/O Connector Type	62-pin D-sub	
Dimensions	167 × 100 mm^2 (6.6 × 3.9 $in.^2$)	
Power Consumption	Typical	3.3 V @ 350 mA, 12 V @ 350 mA
	Max.	3.3 V @ 370 mA, 12 V @ 1000 mA
Temperature	Operating	0~60°C (32~140°F)
	Storage	-40 ~ 70°C (-40 ~ 158°F)
Relative Humidity	Operating	10~90%RH non-condensing
	Storage	5~95%RH non-condensing
Certifications	CE/FCC certified	

Device Installation

For developer, we recommend you to install driver from Advantech Navigator. Advantech Navigator will be installed with DAQNavi SDK Package, so the developer need not download the individual driver package.

The user to build operation environment need not download and install DAQNavi SDK Package, user can only download and install the individual driver Package â DAQNavi_PCIE1824".

- **Install driver for development**
- Ινσταλλ δριϖερ φορ βυιλδινγ οπερατιον ενϖιρονμεντ

Install driver for development:

We offer two ways for driver installation:

- **Installation**
- **Preinstallation**

- Uninstallation

We recommend you use the device auto Installation. We recommend you to insert the card into your system before installing the driver to ensure that the installation process will be completed smoothly.

Installation (Recommended)

You can install the PCIE-1824 card in any PCIE slot on your computer. Please follow the following steps below to install the module on your system.

Step 1: Turn off your computer and unplug the power cord and cables. TURN OFF your computer before installing or removing any components on the computer.

Step 2: Remove the cover of the computer.

Step 3: Remove the slot cover on the back panel of your computer.

Step 4: Touch the metal part on the surface of your computer to neutralize the static electricity that might be on your body.

Step 5: Adjust DIP switch SW1 on board to set the card's board ID.

Step 6: Insert the PCIE-1824 card into a PCIE slot. Hold the card only by its edges and carefully align it with the slot. Insert the card firmly into place. Use of excessive force must be avoided, otherwise the card might be damaged.

Step 7: Fasten the bracket of the PCIE card on the back panel rail of the computer with screws.

Step 8: Connect appropriate accessories(62-pin cable, wiring terminals, etc. if necessary) to the PCIE card.

Step 9: Replace the cover of your computer chassis. Re-connect the cables you removed in step 2.

Step 10: Plug in the power cord and turn on the computer.

Step 11: Install DAQNavi SDK from DVD for building your operation environment.

1. Insert the companion DVD into your DVD disc. The setup program will be launched automatically if you have the auto play function enabled on your system. Then the setup program will be launched..

2. Push the "CONTINUE" button to next step.

3. Select the "Installation" option.

4. Select the "DAQNavi SDK" option to install DAQNavi SDK Package.

5. After DAQNavi SDK have been installed successfully , Advantech Navigator will be lunched automatically. Of course, you can open the Advantech Navigator from "Start>>All Programs>>Advantech Automation>>DAQNavi>>Adventech Navigator" manually. For win7 user, you must run it by right click the item and select the 'Run As Administrator' item from the popup menu.

6. Click "+" on the left of node named "Supported Devices" to expand the contents, then all the supported devices will be listed under this node.

7. Choose the device which you want to install driver, right click on the node, there will be a popup dialog shown, click "Install Driver" on the dialog, thus, the driver will start to install.

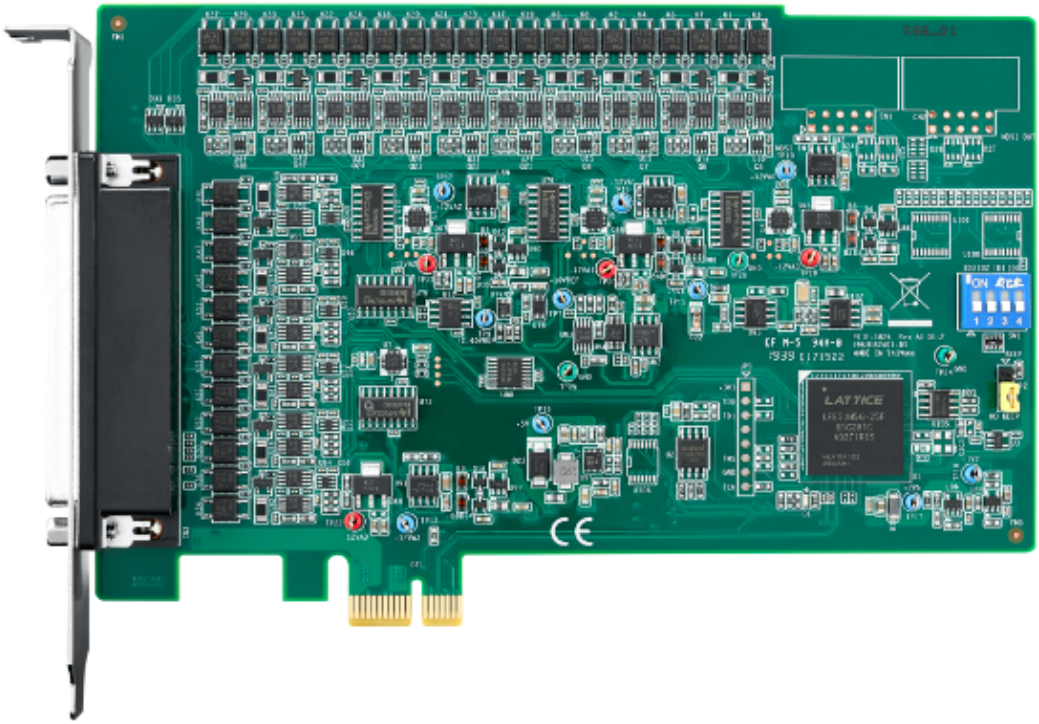
PCI-1756
PCI-1757UP
PCI-1758UDI
PCI-1758UDIO
PCI-1758UDO
PCI-1760/U
PCI-1761
PCI-1762
PCI-1780U
PCI-1784U
PCIE-1730
PCIE-1730H
AIIS-E730
AIIS-1750
PCIE-1744
PCIE-1751
PCIE-1750U
PCIE-1750UH
PCIE-1752
PCIE-1753
PCIE-1754
PCIE-1756
PCIE-1756H
PCIE-1758DIO
PCIE-1758DI
PCIE-1758DO
PCIE-1760
PCIE-1761H
PCIE-1762H
PCIE-1765
PCIE-1802
PCIE-1802L
PCIE-1803
PCIE-1810
PCIE-1812
PCIE-1813
PCIE-1816
PCIE-1816H
PCIE-1824
PCIE-1840
PCIE-1840L

Install Driver

Device Features

PCIE-1824

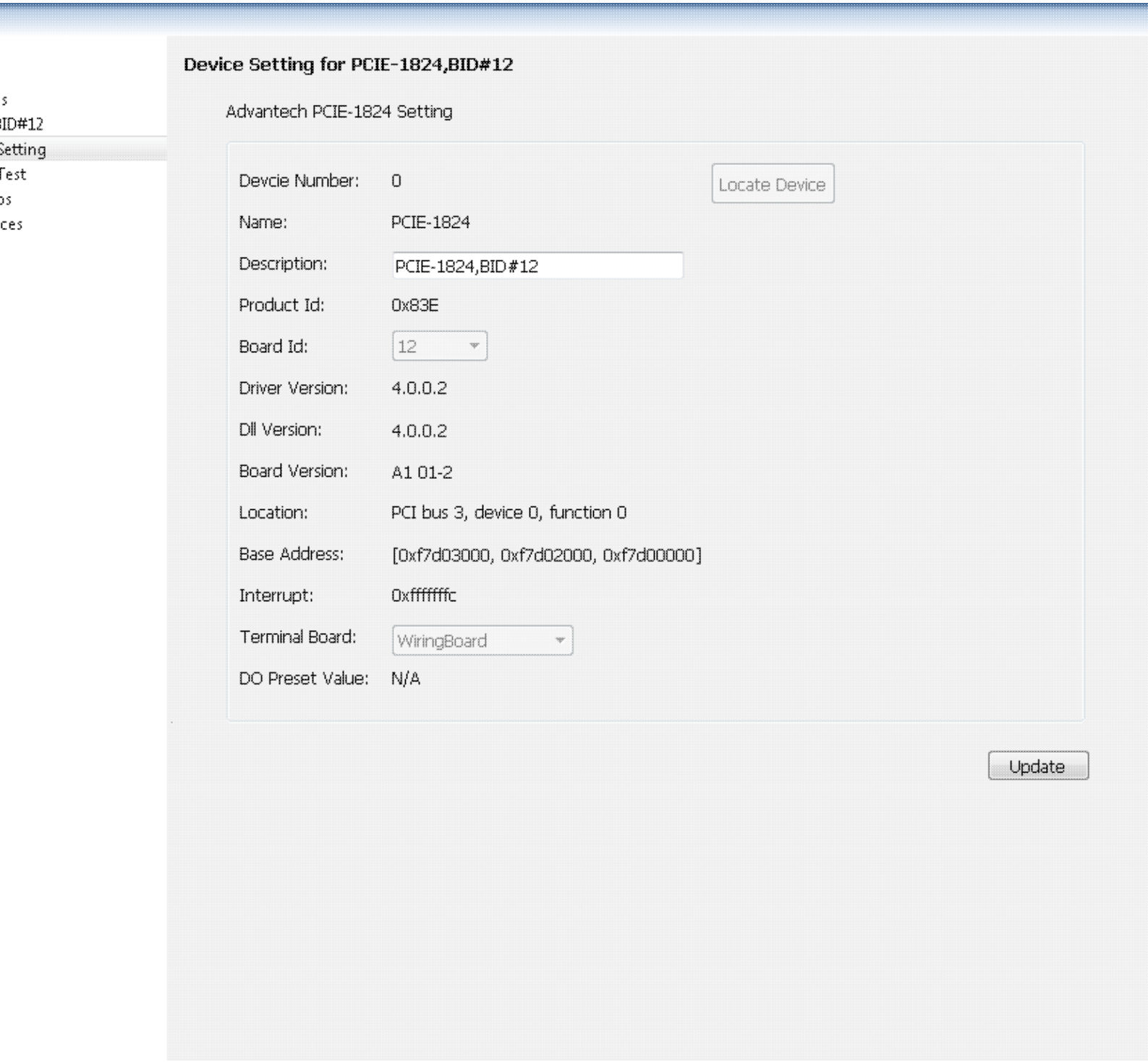
16-bit, 32-ch, high-density multiple channel analog output card for the PCIE bus



Main Features

- 32 high-density analog output channels
- Flexible Output Range: ±10 V, 0 ~ 20 mA and 4 ~ 20 mA
- Synchronized output function
- Keep output values when hot system reset
- High ESD protection (2,000 VDC)
- Board ID switch

8. Click "OK" to finish installation. Navigator detects and installs PCIE-1824 device on the system automatically, then you can find "PCIE-1824" device node under "Installed Devices" node.



Preinstallation

PCIE-1824 DAQNavi Driver supply device preinstallation if you have installed PCIE-1824 DAQNavi Driver from Navigator but without inserting the PCIE-1824 card in your system.

Step 1: Install PCIE-1824 DAQNavi Driver from Navigator.

Step 2: Turn off your computer and unplug the power cord and cables. TURN OFF your computer before installing or removing any components on the computer.

Step 3: Remove the cover of the computer.

Step 4: Remove the slot cover on the back panel of your computer.

Step 5: Touch the metal part on the surface of your computer to neutralize the static electricity that might be on your body.

Step 6: Adjust DIP switch SW1 on board to set the card's board ID.

Step 7: Insert the PCIE-1824 card into a PCIE slot. Hold the card only by its edges and carefully align it with the slot. Insert the card firmly into place. Use of excessive force must be avoided, otherwise the card might be damaged.

Step 8: Fasten the bracket of the PCIE card on the back panel rail of the computer with screws.

Step 9: Connect appropriate accessories to the PCIE card.

Step 10: Replace the cover of your computer chassis. Re-connect the cables you removed in step 2.

Step 11: Plug in the power cord and turn on the computer.

Step 12:

ℜ• In Windows XP system:

Access the "System Device Manager". System will recognize PCIE device and pop up an dialog, click "next" button to complete the installation process. Or else you can install the device driver by right click on the PCIE device item and select "Update" button to install PCIE-1824 driver.

ℜ• In Windows 7 system:

Access the "System Device Manager". System will recognize PCIE-1824 device and the driver of PCIE-1824 will be installed automatically without any notice information.

Note:
If your card has been installed properly, you can see the device name of your card

listed on the "System Device Manager" under "Advantech DAQ device" node. If you do see the device name listed under it but marked with an exclamation sign "!" , it means your device's driver has not been installed correctly and successfully . In this case, you can select the device name and press the "Update" button to update driver manually .

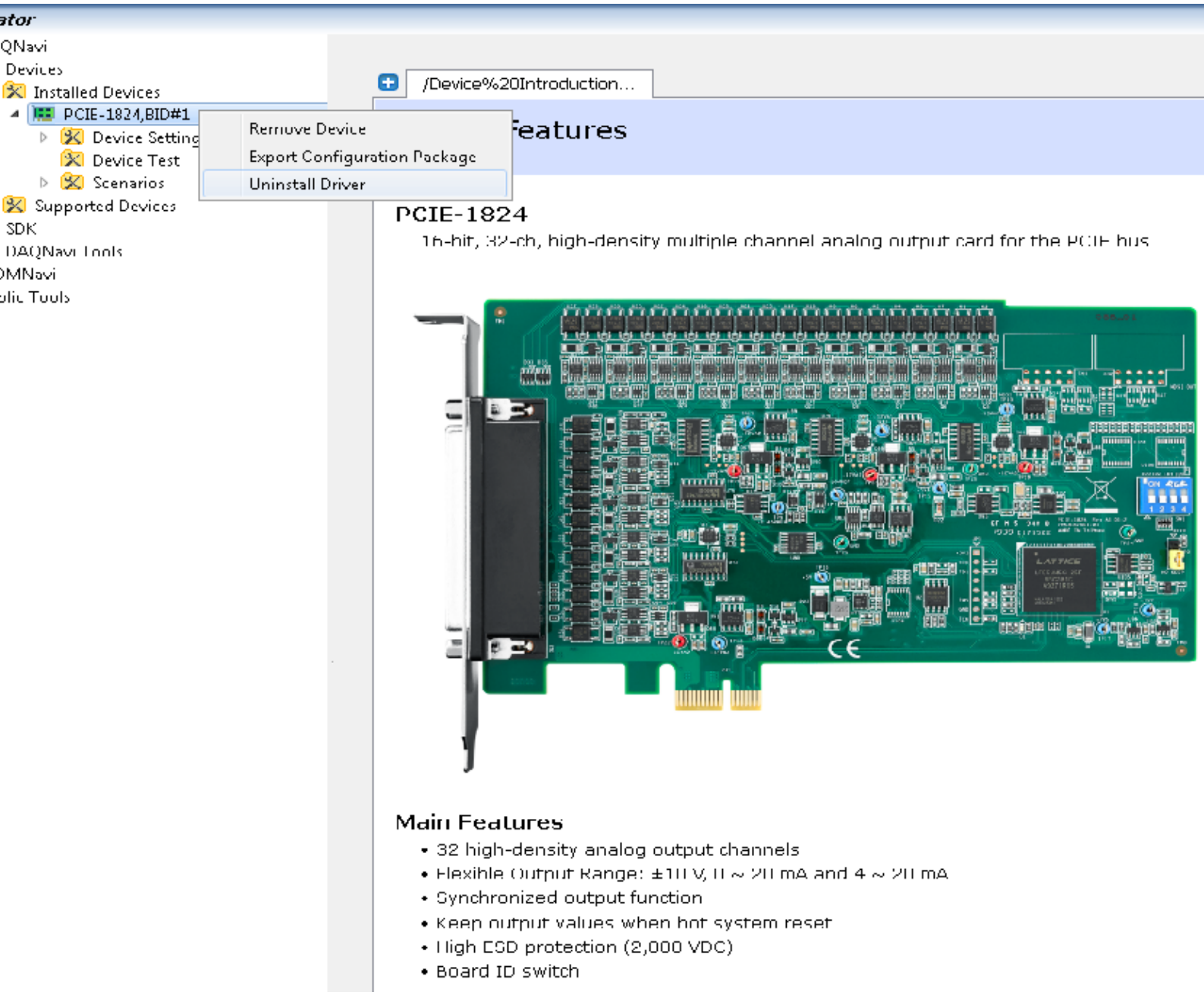
Uninstallation

Step 1: Close the applications of the PCIE-1824.

Step 2: Run Advantech Navigator. For win7 user, you must run it by right click the item and select the 'Run As Administrator' item from the popup menu.

Step 3: Click "+" on the left of node named "Installed Devices" to expand the contents, then all the installed devices will be listed under this node.

Step 4: Choose the device which you want to uninstall driver, right click on the node, there will be a popup dialog shown, click "uninstall Driver" on the dialog, thus, the driver will start to uninstall.


QNavi
Devices
Installed Devices
PCIE-1824,BID#1
Device Setting
Device Test
Scenarios
Supported Devices
SDK
DAQNavi Tools
DMNavi
Remove Device
Export Configuration Package
Uninstall Driver
/Device%20Introduction...
Features
PCIE-1824
16-bit, 32-ch, high-density multiple channel analog output card for the PCIE bus
Main Features
• 32 high-density analog output channels
• Flexible Output Range: ±10 V, 0 ~ 20 mA and 4 ~ 20 mA
• Synchronized output function
• Keep output values when hot system reset
• High ESD protection (2,000 VDC)
• Board ID switch

Install driver for building operation environment:

We offer two ways for driver installation:

- **Installation**
- **Preinstallation**
- **Uninstallation**

We recommend use the driver auto Installation way. We recommend you to install the card into your system before installing the driver, thus to ensure that the device driver will be installed smoothly.

Installation (Recommended)

You can install the PCIE-1824 card in any PCIE slot on your computer. Please follow the steps below to install the module on your system.

Step 1: Turn off your computer and unplug the power cord and cables. TURN OFF your computer before installing or removing any components on the computer.

Step 2: Remove the cover of the computer.

Step 3: Remove the slot cover on the back panel of your computer.

Step 4: Touch the metal part on the surface of your computer to neutralize the static electricity that might be on your body.

Step 5: Adjust DIP switch SW1 on board to set the card's board ID.

Step 6: Insert the PCIE-1824 card into a PCIE slot. Hold the card only by its edges and carefully align it with the slot. Insert the card firmly into place. Use of excessive force must be avoided, otherwise the card might be damaged.

Step 7: Fasten the bracket of the PCIE card on the back panel rail of the computer with screws.

Step 8: Connect appropriate accessories(62-pin cable, wiring terminals, etc. if necessary) to the PCIE card.

Step 9: Replace the cover of your computer chassis. Re-connect the cables you removed in step 2.

Step 10: Plug in the power cord and turn on the computer.

Step 11: Install PCIE-1824 DAQNavi driver from DVD for building your operation environment.

1. Insert the companion DVD into your DVD disc. The Setup program will be launched automatically if the auto play function enabled on your system. When the Setup Program is launched, you will see the following setup screen.

2. push the "CONTINUE" button to next step.

3. Select the "Installation" option.

4. Select the "Install Windows Individual Driver" option.

5. Select the "PCIE Series" option.

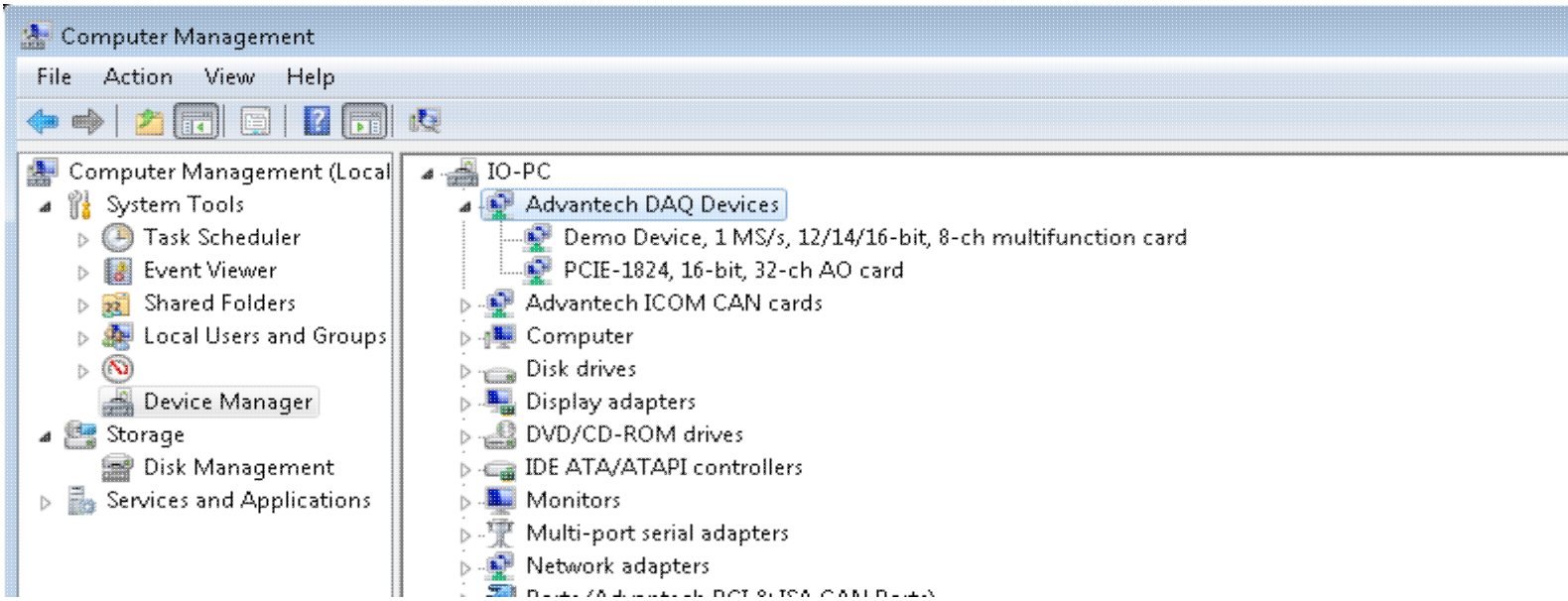
6. Select the "PCIE-1824" option to run the PCIE-1824 DAQNavi Driver package.

7. Click "Finish". The installer detects and installs PCIE-1824 device on the system automatically. The device driver files and manual can be found at: System disk\Advantech\DAQNavi\Driver\PCIE1824.

Step 12: After the PCIE-1824 driver is installed, you can verify whether it is properly installed on your system in the System Device Manager:

1. Access the System Device Manager.

2. The device name of the PCIE-1824 should be listed under the Advantech DAQ Devices tab.



Preinstallation

PCIE-1824 DAQNavi Driver supply device preinstallation if you have installed PCIE-1824 DAQNavi Driver from DVD but without inserting the PCIE-1824 card in your system.

Step 1: Install PCIE-1824 DAQNavi Driver from DVD

Step 2: Turn off your computer and unplug the power cord and cables. TURN OFF your computer before installing or removing any components on the computer.

Step 3: Remove the cover of the computer.

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Step 9: Connect appropriate accessories to the PCIE card.

Step 10: Replace the cover of your computer chassis. Re-connect the cables you removed in step 2.

Step 11: Plug in the power cord and turn on the computer.

Step 12:

- In Windows XP system:

  Access the "System Device Manager". System will recognize PCIE device and pop up an dialog, click "next" button to complete the installation process. Or else you can install the device driver by right click on the PCIE device item and select "Update" button to install PCIE-1824 driver.

- In Windows 7 system:

Access the "System Device Manager". System will recognize PCIE-1824 device and the driver of PCIE-1824 will be installed automatically without any notice information.

Note: If your card has been installed properly, you can see the device name of your card listed on the "System Device Manager" under "Advantech DAQ device" node. If you do see the device name listed under it but marked with an "!" sign, it means your device's driver has not been installed correctly and successfully . In this case, you can

select the device name and press the "Update" button to update driver manually.

Uninstallation

Step 1: Close the applications of the PCIE module.
Step 2: Run the PCIE-1824 DAQNavi Driver package, select the "Remove", then the PCIE-1824 driver will be uninstalled.

After your card is properly installed on your system, you can now configure your device using the Device Configuration Program that has itself already been installed during driver installation.

The DAQNavi Driver provides device setting dialog box that allows you to configure your device, and later stores your settings on the system registry. These settings will be used when you call the DAQNavi SDK to manipulate functions of device. Device Configuration assists to use Advantech DAQ cards more efficiently and easily.

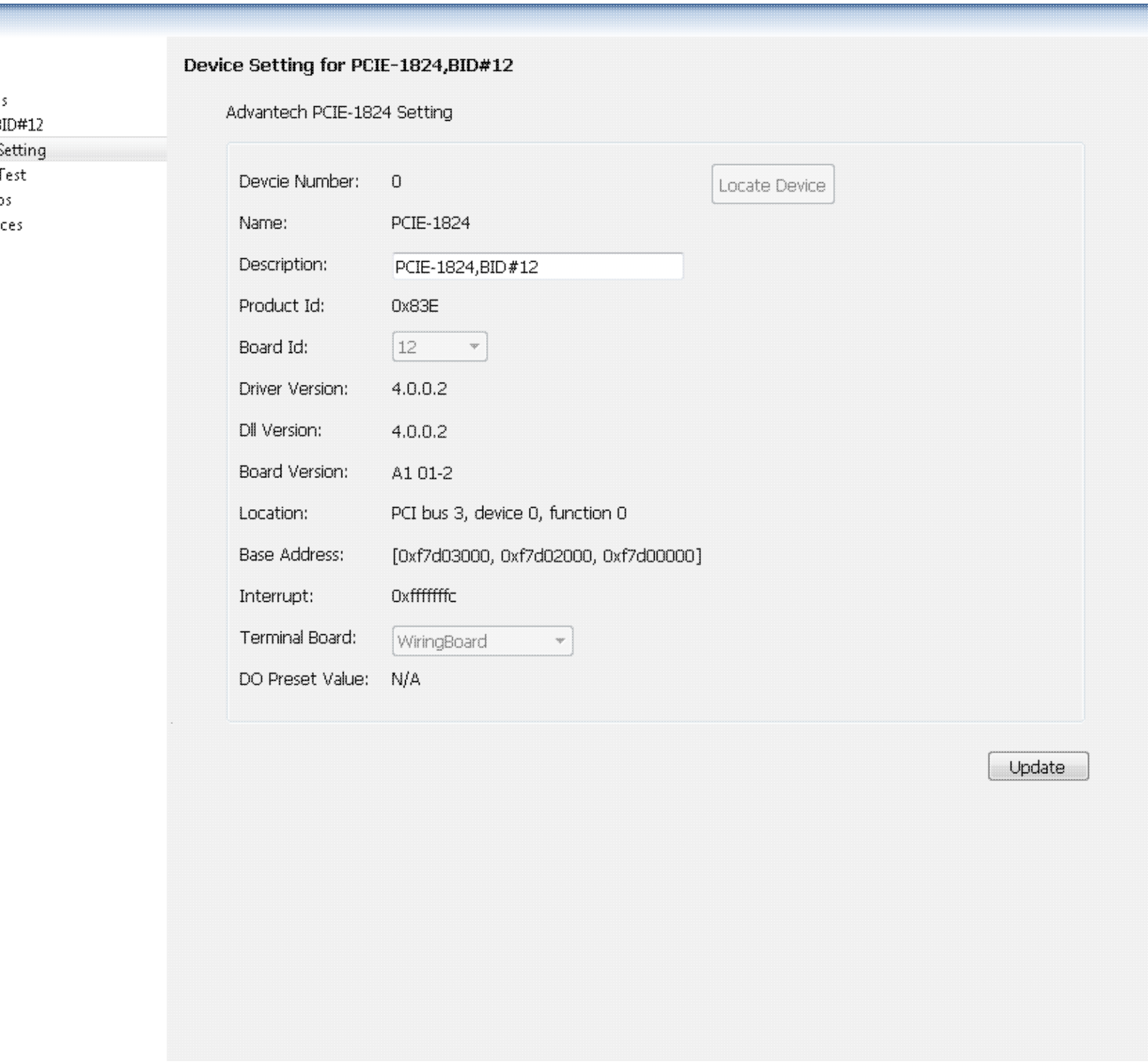
Configuring the Device:

After DAQNavi SDK installation, you can access Advantech Navigator from "Start Â» Programs Â» Advantech Automation Â» DAQNavi Â» Advantech Navigator" and open Navigator window.

Click the "+" on the left to unfold the content. If the software and hardware installation are completed, the PCIE-1824 is listed under DAQNavi Â» Devices Â» Installed Devices.

1. Update device properties

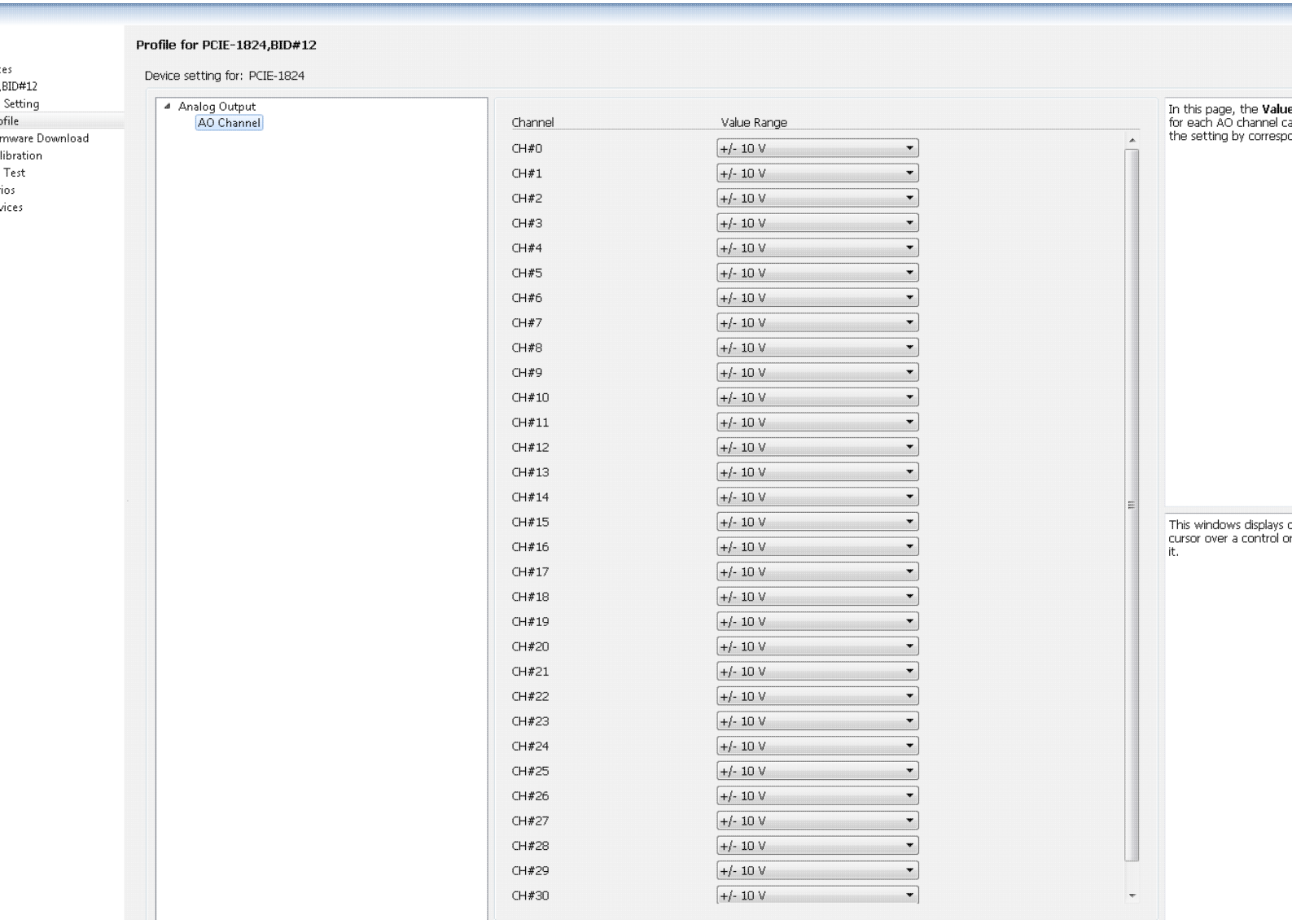
Select Device Setting to open PCIE-1824 device setting dialog box on the right.


Device Setting for PCIE-1824,BID#12
Advantech PCIE-1824 Setting
Devcie Number: 0
Locate Device
Name: PCIE-1824
Description: PCIE-1824,BID#12
Product Id: 0x83E
Board Id: 12
Driver Version: 4.0.0.2
Dll Version: 4.0.0.2
Board Version: A1 01-2
Location: PCI bus 3, device 0, function 0
Base Address: [0xf7d03000, 0xf7d02000, 0xf7d00000]
Interrupt: 0xfffffffc
Terminal Board: WiringBoard
DO Preset Value: N/A
Update

On the device setting dialog box, you can change default settings of Device function, and click the "Update" button to apply the device setting to the device.

2. Configure device function

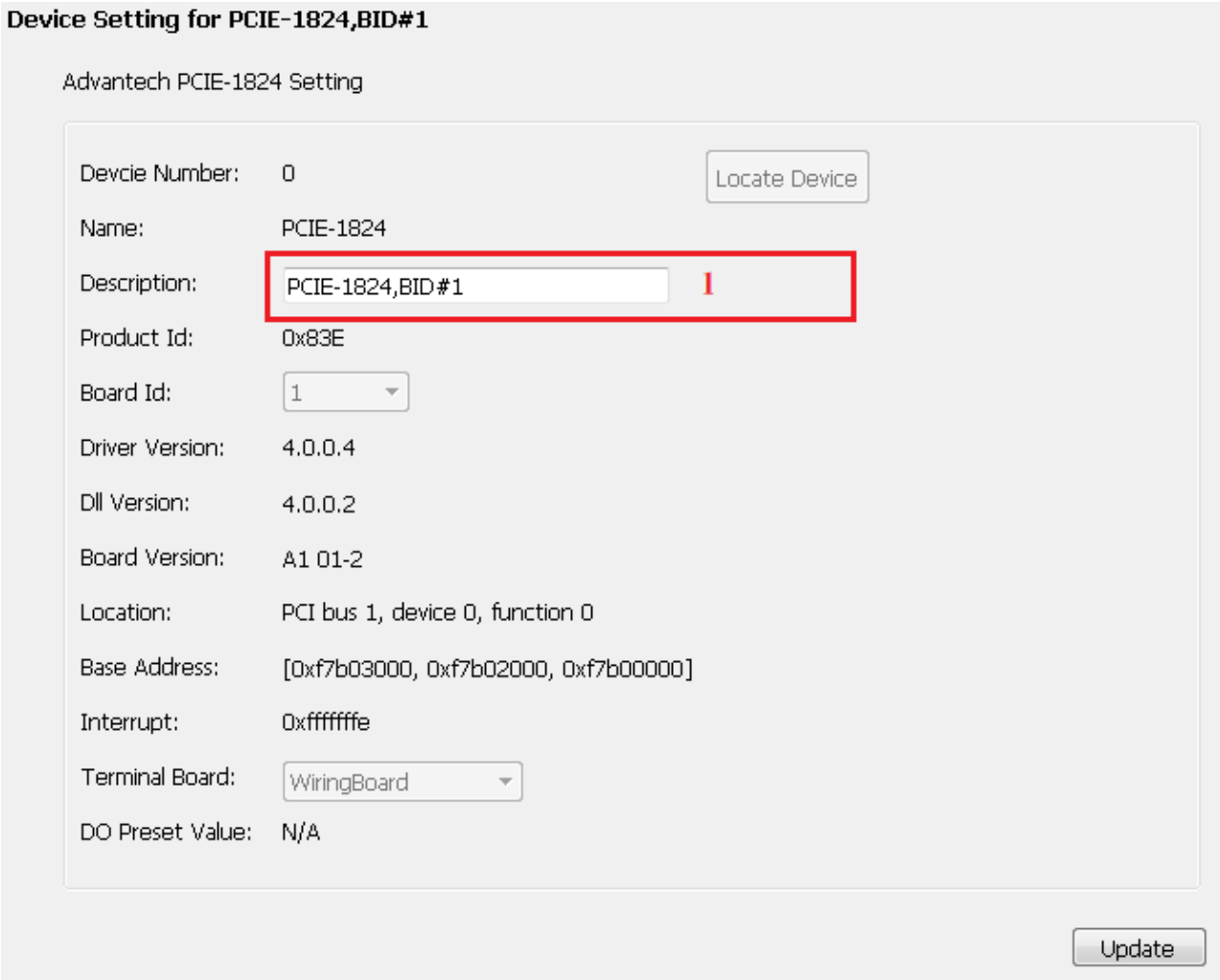
Select Profile to open PCIE-1824 profile dialog box on the right.



On the profile dialog box, you can change default settings of Analog Output function,

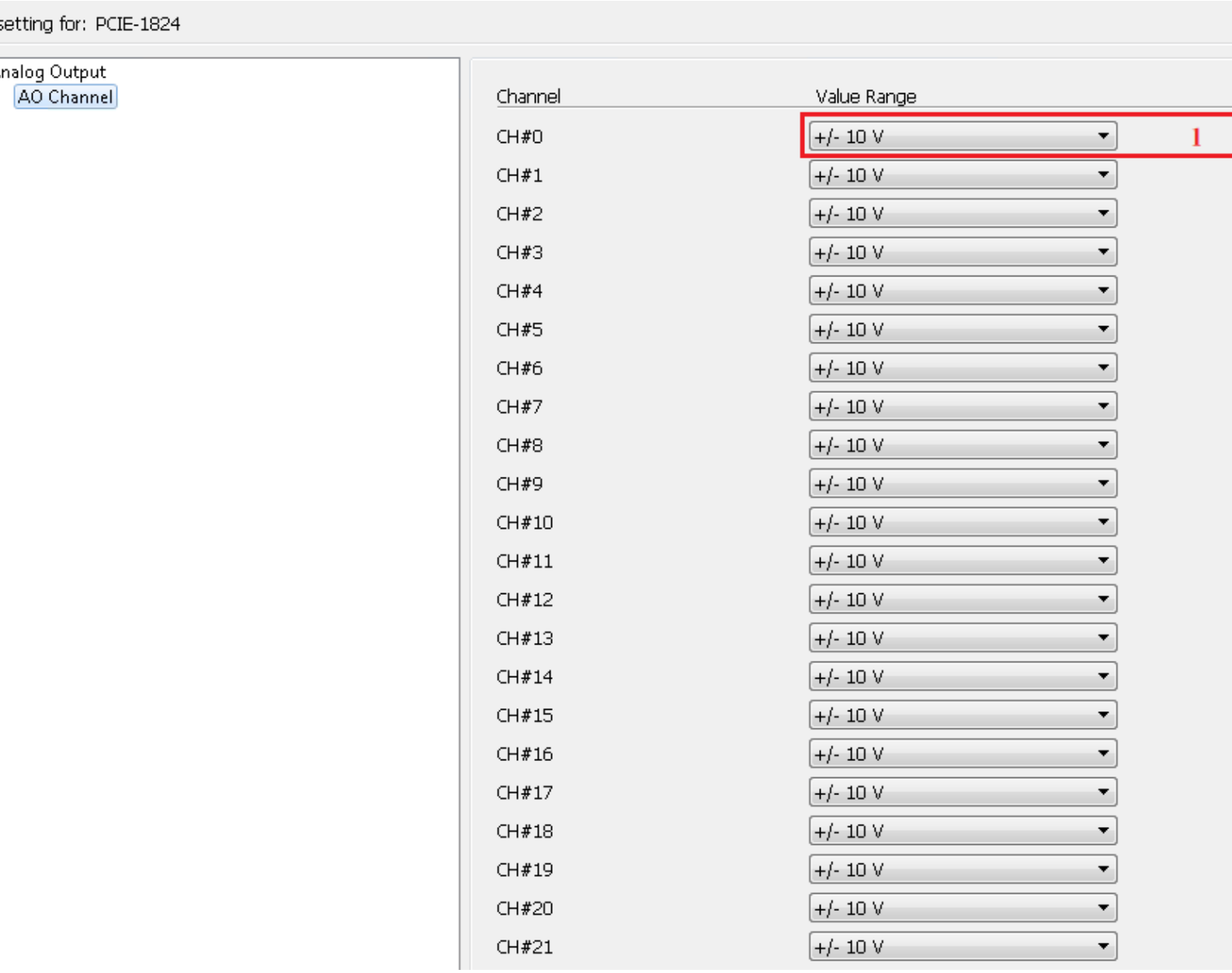
With the corresponding check box checked, you can click the "Save" button to store current configuration into the profile, and you can use this file to configure the device.

Device Configuration: Select "**Device Setting**" tab to configure device.



Item 1: Set device descriptions.

Analog Output Configuration: Select "**Analog Output**" tab to configure analog output.



Item 1: Select ComboBox to set reference voltage of each channel.

Note:
The PCIE-1824 has been calibrated at the factory for initial use. You are not required to calibrate the PCIE-1824 in normal conditions.

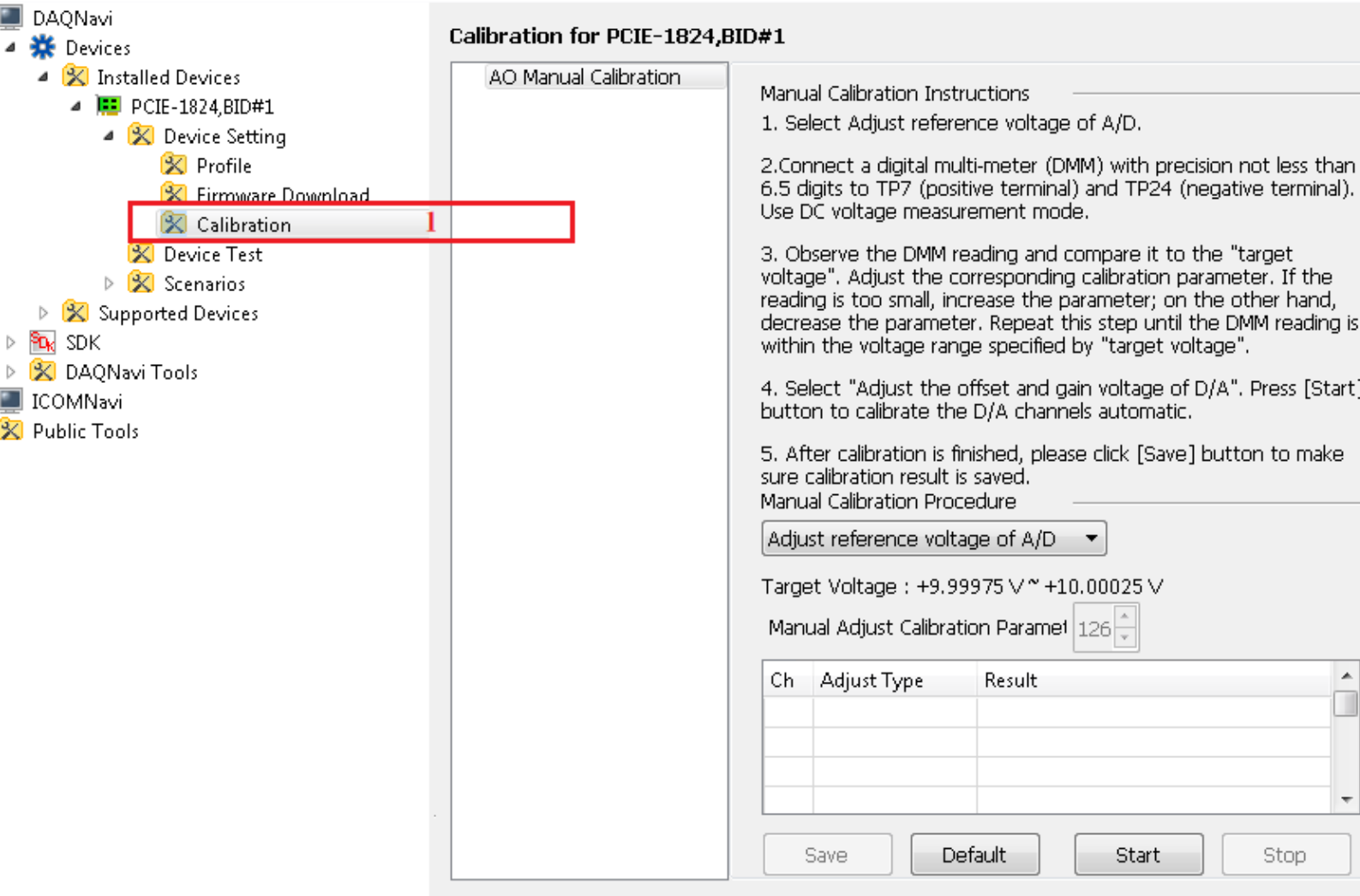
This section provides brief information on PCIE-1824 calibration. Regular calibration checks are important to maintain accuracy in data acquisition and control application.

To perform a satisfactory calibration, users need a 6-1/2 digit stable for the calibration process. Because difficult to get high-accuracy standard DC voltage source, so we only offered A/D auto calibration way.

PCIE-1824 Setting Calibration provides functions :

- Analog Output Calibration

Please Click the "Calibration" tab in configuration dialog box and follow the calibration instructions to finish your calibration.

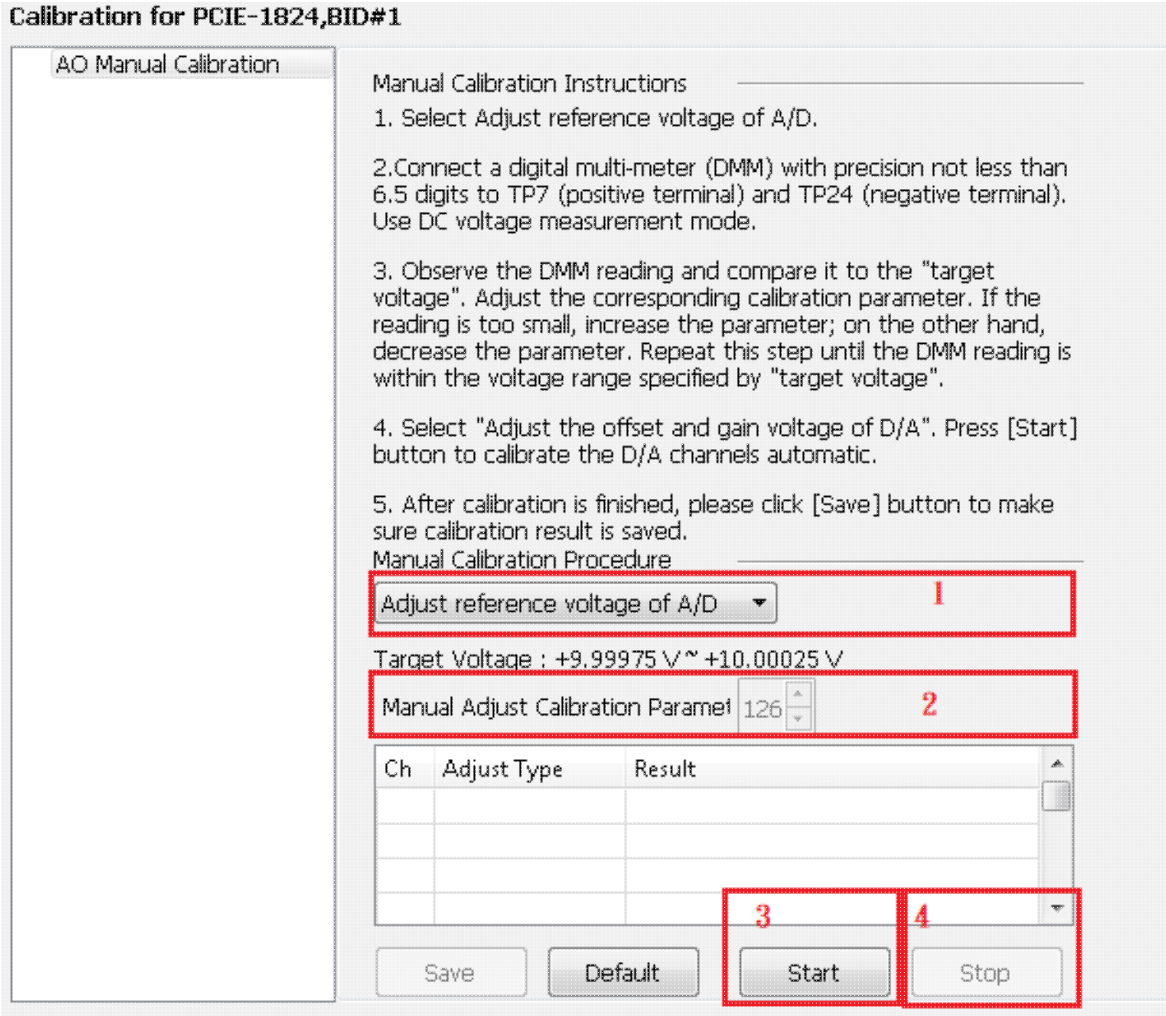


Item1 :Display Analog Output Manual Calibration.

1. Analog Output Calibration

- **Adjust the Reference voltage**

Please adjust the reference voltage before you start to analog calibration.



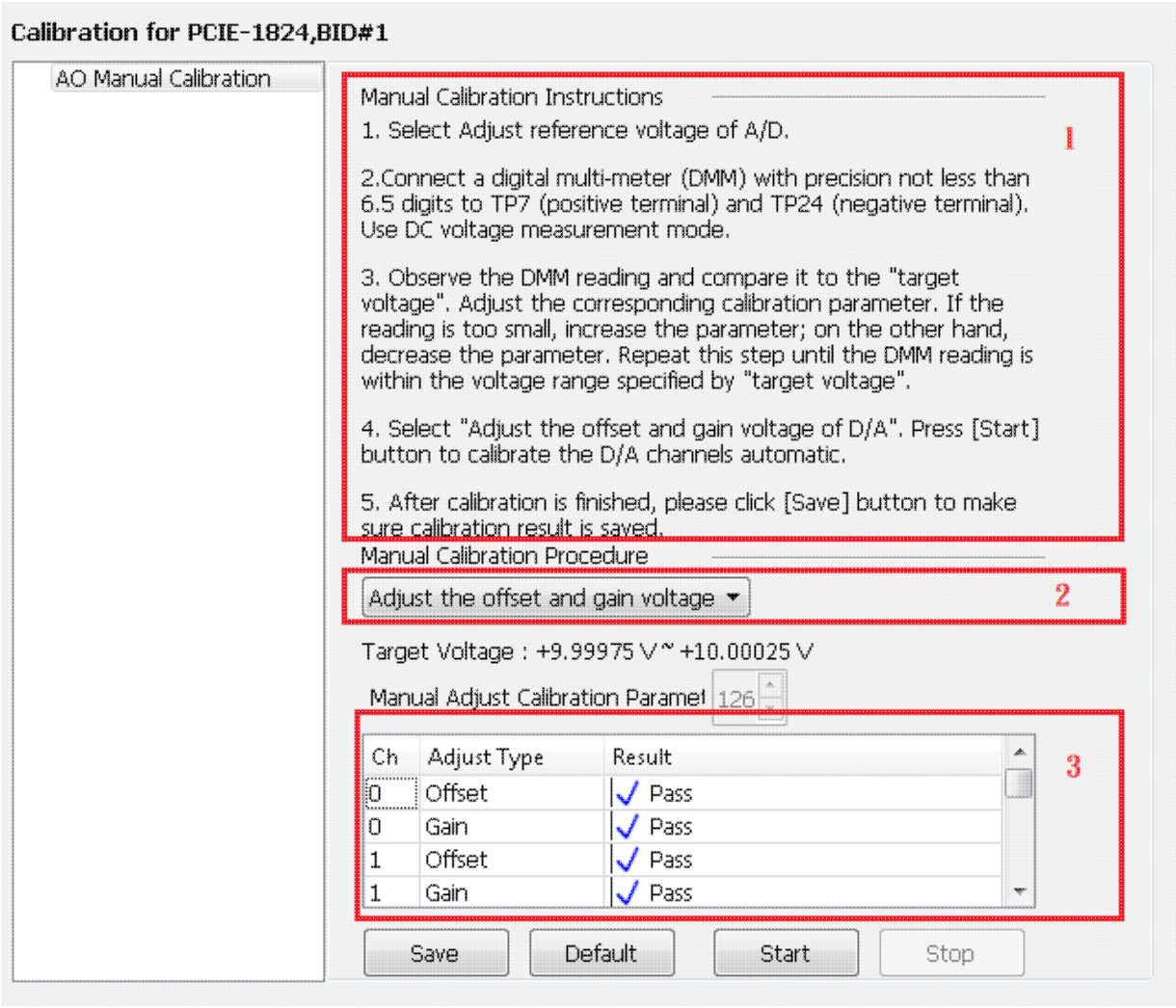
Step 1 : Select the calibration item is 'Adjust Reference Voltage of A/D'.

Step 2 : Press [Start] button.

Step 3 : Click the 'Manual Adjust Calibration Parameter' until the voltage on TP7 and TP24 has reached Target Voltage.

Step 4 : Press [Stop] button to stop the adjusting.

- **Analog Output calibration**



Item 1 : Display the Calibration Instructions.

Item 2 : The Calibration item.

Item 3 : Display the calibration process.

Notice : Before starting the auto calibration procedure, Please ensure reference voltage on TP7 and TP24 is reached Target Voltage.

Press start button, the calibration procedure will be auto run until calibration finished.

After finished calibration, please click 'Save' button to make sure calibration result is saved.

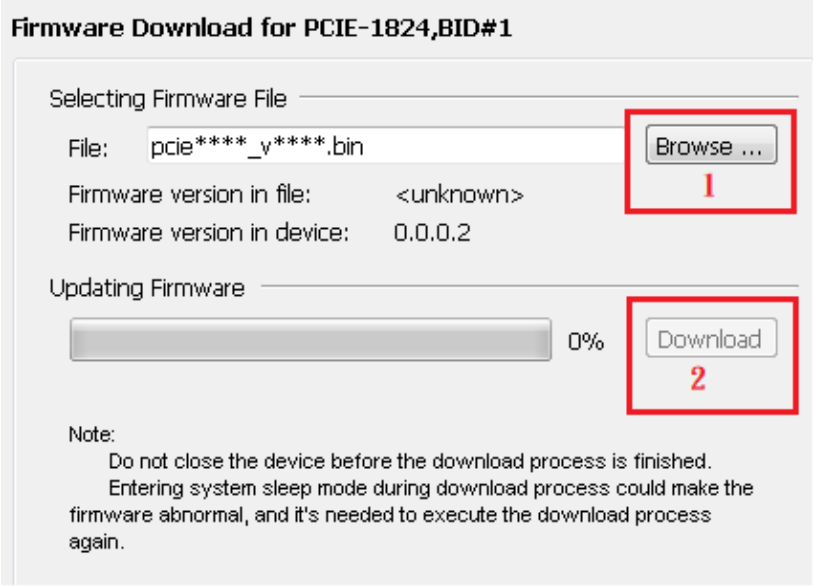
Step 1 : Select the calibration item is 'Adjust the offset and gain Voltage'.

Step 2 : Press [Start] button.

Step 3 : After calibration is finished, If all AO channels are passed, please click [Save] button to make sure calibration result is saved.

Otherwise, Press [Start] button calibrate again.

Update Firmware: Select "Firmware Download" tab to load firmware.



Step 1 : Click "Browse" button in Item 1 to open file dialog. Select the .bin firmware file.

Step 2 : Click "Download" button in Item 2 to update firmware of PCIE-1824.

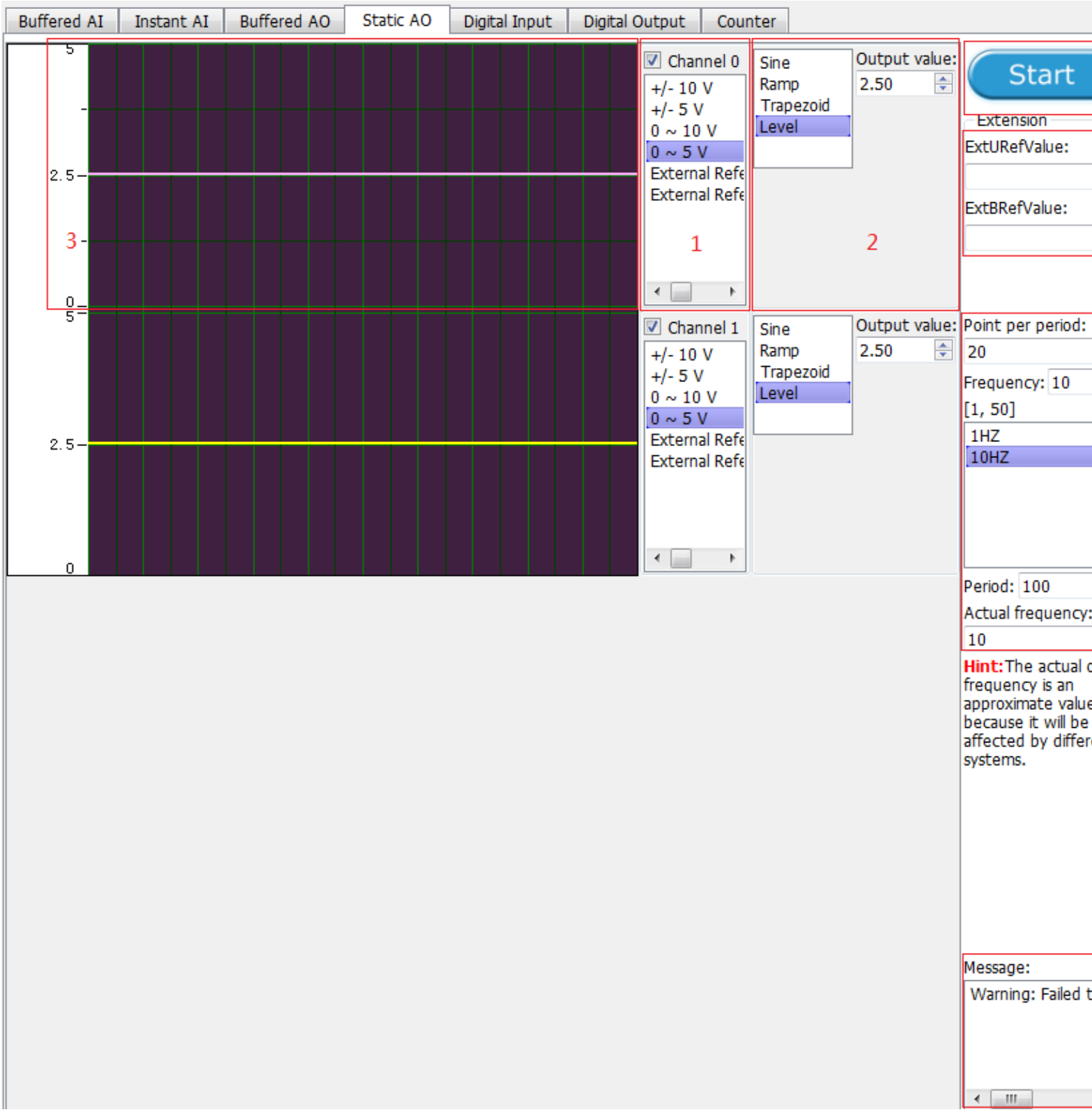
After device setup, you can select Device Test to open PCIE-1824 device test dialog box in Navigator window.

In the "Device Test" dialog box, users can select different tabs to test various functions of PCIE-1824.

- Analog Output Test

1. Static AO Test

Click the "Static AO" tab in the "Device Test" interface. All the AO physical channels of the device will be listed on the right.



Item 1: Shows the number of a corresponding analog output channel. Check the checkbox and the corresponding analog output channel will be selected.

The list box list value range type for each channel. The user can select the value range type in the list box.

Item 2: Shows the waveforms to be output analog output channel, and check domain Up Down button to set amplitude and DC offset value of waveform.

Item 3: Shows analog output waveform in the graph.

Item 4: Set external reference voltage value of AO.

Item 5: Set the point per period and frequency of waveform, and list box is also used to adjust frequency of waveform.

Item 6: "Start" button is used to start analog output.

DAQNavi SDK has a series of application programming interfaces, which is a library of classes, methods, and properties for creating applications for your device. For information about how to program with DAQNavi SDK, please refer to DAQNavi Interface Manual Programming with CSCL.

Property List

Device properties are stored in the system and read by the driver when it is loaded, and then the device is programmed according to these loaded values. Different devices have different properties because of the different hardware they use. Property in DAQNavi can be divided into two category, one is named **Features**, the other is truly property. **Features** property is description of the device characteristics,which is read only, you can get it's values but can't set value for these features, so in DAQNavi, you will find the features only has Get method. Property can be set or got by programming, some properties also can be set by the configuration dialog box in device manger. you can set value to modify the writable property of the device in running time.

Following table is a property list supported by PCIE-1824, the value of these property and only lists the default value for the writable properties. User can modify writable property value according to corresponding features.

Property supported by PCIE-1824:

			ã□□Property	Default Value	Available Value
Analog Output	AoCtrlBase		ChannelCount	32	
			Channels	An array of AoChannel.	
		Features	Resolution		16
			DataSize		4
			DataMask		0xffff0000
			ChannelCountMax		31
			ValueRanges		{V_Neg10To10, mA_0To20, mA_4To20}
			ExternalRefAntiPolar		false
			BufferedAoSupported		false

Method List

The DAQNavi SDK Class Library provides a series of methods to manipulate DAQ device functions . Following is an method list which supported by the device PCIE-1824.

Methods supported by PCIE-1824 :

		ã□□
Analog Output	**InstantAoCtrl**	**Write**

Event ID List

Following is an Event list which supported by the device PCIE-1824.

Event supported by PCIE-1824:

Device	**DeviceCtrl**	**PropertyChanged**	**This event occurs when the low-level device's properties changing. It is only effective for the device which selected with 'ModeRead'. you can reload the latest property value by 'UpdateProperties' method.**